Relaxation and Odd-Even Effect in Fermi Liquids

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Outline

1. **Introduction**
   - Boltzmann equation
   - Fermi Dirac distribution
2. **Recovering equations**
   - Linearization of the collision integral
   - Eigenvalue problem
3. **Odd-Even effect**
4. **Further works**

1 Introduction

1.1 Boltzmann equation

Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}}\right) f(t, \mathbf{x}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{x}, \mathbf{p})] \quad (1)$$

This talk follows the treatment of the linearized collision integral in Nilsson et al. [1], which builds on the mode hierarchy first identified in Ledwith et al. [2].

$f(t, \mathbf{x}, \mathbf{p})$: One-particle density function

$\frac{\partial f}{\partial t}$: Local time evolution

$\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}}$: Streaming/advection term

$\mathbf{F} \cdot \frac{\partial f}{\partial \hbar \mathbf{p}}$: Force/acceleration term due to external fields

$\hat{\mathcal{J}}[f]$: Collision integral - effect of collisions on distribution function

1.2 Fermi Dirac distribution

Fermi-Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1} \tag{2}$$

- $\beta = 1/k_B T$ is the inverse temperature
- $\varepsilon(\mathbf{p}) = \dfrac{\hbar^2 \mathbf{p}^2}{2m}$ is the energy of particles with momentum $\mathbf{p}$
- μ is the chemical potential

The Fermi-Dirac distribution is in **equilibrium**.

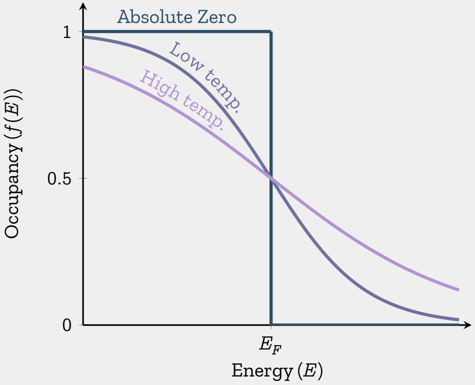

Figure 1: Fermi-Dirac distribution at different temperatures

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

k-space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k-space corresponds to a wave $\mathrm{e}^{\mathrm{i}\mathbf{k}}$
- **Fourier transform** converts between k-space to real space

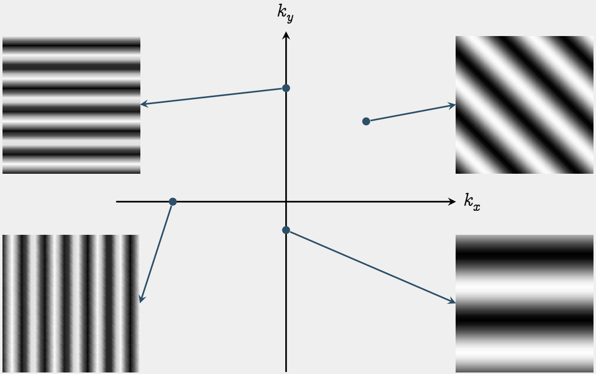

Figure 2: Representative waves of k-space

- Distance from origin $\propto$ Wave frequency
- Direction on k-space $\implies$ Wave direction

Fermi surface

- Surface in k-space that separates *occupied electron states* from *unoccupied* electron states
- Determined by periodicity and symmetry of crystal lattice.
- Changes with temperature.

Collision integral

$$\hat{\mathcal{J}}[f] = -\int \dots \int \frac{1}{(2\pi)^6} \, \mathrm{d}\mathbf{p}_2 \, \mathrm{d}\mathbf{p}_1' \, \mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B}, \tag{3}$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'})(1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1)(1 - f_2), \tag{4}$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \tag{5}$$

$W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2)$: Collision rates

$\mathcal{B}$: "Product of Fermi factors"[1]

[1]Non-standard in literatures

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}}\right) f(t, \mathbf{x}, \mathbf{p}) = -\int \dots \int \frac{1}{(2\pi)^6} \, \mathrm{d}\mathbf{p}_2 \, \mathrm{d}\mathbf{p}_1' \, \mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \quad (6)$$

- Very hard (might be impossible) to solve analytically
- Non-linear

Ideas of attacking the equation

- Consider only perturbations near equilibrium: $f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$
- Linearize $\hat{\mathcal{J}}[f]$
- Let there be no external force and no thermal gradient

We will end up with

$$\hat{\mathcal{J}}[\phi] = -\gamma\phi \tag{7}$$

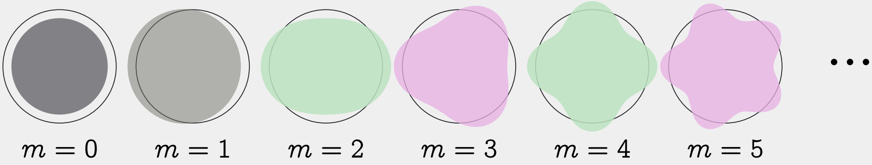

Figure 3: Circular harmonical perturbations of the Fermi surface

1. Perturbation is smooth
2. Collision integral is *linear*

The perturbation can be expanded in the harmonics basis: $e^{im\theta}$.

2 Recovering equations

2.1 Linearization of the collision integral

Preparation

Fermi equilibrium: $f_0(\mathbf{p}) = \dfrac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \quad (8)$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \quad (9)$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \quad (10)$$

$$f(t,\mathbf{p}) = f_0(\mathbf{p}) + \delta f(t,\mathbf{p}) \qquad \text{Perturbation} \tag{11}$$

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu}\,\delta\mu \qquad \text{Taylor series. Fix } \varepsilon \tag{12}$$

$$\therefore \delta f(t,\mathbf{p}) = \frac{\partial f_0}{\partial \mu}\,\delta\mu \tag{13}$$

> **Perturbation of interest:** $\delta\mu = T\psi$ (temperature dependent change in chemical potential)

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T\psi \qquad (14)$$

$$= \left(-\frac{1}{T} f_0(1 - f_0)\right) \times T\psi \qquad (15)$$

$$= f_0(1 - f_0)\psi \qquad (16)$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0:

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \tag{17}$$

The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \tag{18}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}), \tag{19}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1}f_{0,2}(1 - f_{0,1}), \tag{20}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,2'}) - f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}), \tag{21}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,1'}) - f_{0,1'}(1 - f_{0,1})(1 - f_{0,2}). \tag{22}$$

Substitute $\delta f_i = f_{0,i}(1 - f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \Big[f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'} f_{0,2'}(1 - f_{0,2})\Big] \times f_{0,1}(1 - f_{0,1})\psi_1 \tag{23}$$

$$= \Big[f_{0,1} f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})(1 - f_{0,1}) + f_{0,1} f_{0,1'} f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})\Big]\psi_1$$

$$= \Big(f_{0,1} f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})\Big)\Big[(1 - f_{0,1}) + f_{0,1}\Big]\psi_1 \tag{24}$$

$$= \Big(f_{0,1} f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})\Big)\psi_1 \tag{25}$$

Define $F_{121'2'} = f_{0,1} f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) = f_{0,1'} f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'}\psi_1 \tag{26}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'} \psi_i \qquad (27)$$

Hence,

$$\begin{aligned} \mathcal{B}_{\mathrm{lin}} &= \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} && (28) \\ &= F_{121'2'} \left[\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}\right] && (29) \\ &= F_{121'2'} \left[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\right] && (30) \end{aligned}$$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}'_2 W(\mathbf{p}'_1,\mathbf{p}'_2|\mathbf{p}_1,\mathbf{p}_2)\mathcal{B} \tag{31}$$

$$= -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}'_2 W(\mathbf{p}'_1,\mathbf{p}'_2|\mathbf{p}_1,\mathbf{p}_2)(\mathcal{B}_{f_0}+\mathcal{B}_{\text{lin}}+\mathcal{B}_{\text{non-lin}}) \tag{32}$$

- Collision integral vanishes at equilibrium: $\hat{\mathcal{J}}[f_0] = 0$
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}'_2 \times W(\mathbf{p}'_1,\mathbf{p}'_2|\mathbf{p}_1,\mathbf{p}_2)F_{121'2'}[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}'_1)-\psi(\mathbf{p}'_2)] \tag{33}$$

Expression of $F_{121'2'}$

- For convenience, write $F_{121'2'}$ as **total momentum** (P) and **center of momentum** (Q)

$$\mathrm{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathrm{Q} = \frac{1}{2}\mathbf{p}_1 - \frac{1}{2}\mathbf{p}_2. \tag{34}$$

$$\mathbf{p}_1 = \frac{1}{2}\mathbf{P} - \mathbf{Q}, \tag{35}$$

$$\mathbf{p}_2 = \frac{1}{2}\mathbf{P} + \mathbf{Q}, \tag{36}$$

$$\mathbf{P}\cdot\mathbf{Q} = \frac{1}{2}\left(p_1^2 - p_2^2\right). \tag{37}$$

$$\varepsilon(\mathbf{p}_1) = \frac{\hbar^2 p_1^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 - \mathbf{P}\cdot\mathbf{Q}\right), \tag{38}$$

$$\varepsilon(\mathbf{p}_2) = \frac{\hbar^2 p_2^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 + \mathbf{P}\cdot\mathbf{Q}\right). \tag{39}$$

Define[2]

$$\xi = \frac{\hbar^2 \beta}{2m^*}, \quad \text{and} \quad X = \beta\left(\frac{\hbar^2 P^2}{8m^*} + \frac{\hbar^2 Q^2}{2m^*} - \mu\right) \tag{40}$$

Then,

$$\beta\big(\varepsilon(\mathbf{p}_1) - \mu\big) = \beta\left(\frac{\hbar^2}{8m^*}P^2 + \frac{\hbar^2}{2m^*}Q^2 - \mu\right) - \frac{\hbar^2 \beta}{2m^*}\mathbf{P}\cdot\mathbf{Q} \tag{41}$$

$$= X - \xi \mathbf{P}\cdot\mathbf{Q}. \tag{42}$$

Similarly,

$$\beta\big(\varepsilon(\mathbf{p}_2) - \mu\big) = X + \xi \mathbf{P}\cdot\mathbf{Q}. \tag{43}$$

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[2] X can possibly be interpretted as: *average deviation from background chemical potential*

Hence,

$$f_0(\mathbf{p}_1) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}]}. \tag{44}$$

$$f_0(\mathbf{p}_2) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \tag{45}$$

Recall that $F_{121'2'} = f_0(\mathbf{p}_1)f_0(\mathbf{p}_2)(1 - f_0(\mathbf{p}_1'))(1 - f_0(\mathbf{p}_2'))$

$$f_0(\mathbf{p}_1)f_0(\mathbf{p}_2) = \frac{1}{\exp[X - \xi \mathrm{P} \cdot \mathrm{Q}] \exp[X + \xi \mathrm{P} \cdot \mathrm{Q}]} \tag{46}$$

$$= \frac{1}{\exp[2X] + \exp[X](\exp[\xi \mathrm{P} \cdot \mathrm{Q}] + \exp[-\xi \mathrm{P} \cdot \mathrm{Q}]) + 1} \tag{47}$$

$$= \frac{1}{\exp[2X] + 2\exp[X]\cosh(\xi \mathrm{P} \cdot \mathrm{Q}) + 1} \tag{48}$$

$$= \frac{1}{2\left(\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})\right)} \tag{49}$$

By symmetry,

$$(1 - f_0(\mathbf{p}_1'))(1 - f_0(\mathbf{p}_2')) = \frac{1}{2\left(\cosh X + \cosh(\xi \mathrm{P}' \cdot \mathrm{Q}')\right)} \tag{50}$$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})][\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})]} \tag{51}$$

Expression of W - the collision rate

$$W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2) = \frac{2\pi}{\hbar}|V|^2 \times (2\pi)^2 \times \delta(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_1'-\mathbf{p}_2')\delta\left(\varepsilon(\mathbf{p}_1)+\varepsilon(\mathbf{p}_2)-\varepsilon(\mathbf{p}_1')-\varepsilon(\mathbf{p}_2')\right) \quad (52)$$

$$\hat{\mathcal{J}}[f(t,\mathbf{p}_1)] = -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2' \times \frac{2\pi}{\hbar}|V|^2 \times (2\pi)^2 \times \delta(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_1'-\mathbf{p}_2')\delta\left(\varepsilon(\mathbf{p}_1)+\varepsilon(\mathbf{p}_2)-\varepsilon(\mathbf{p}_1')-\varepsilon(\mathbf{p}_2')\right) \times F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \quad (53)$$

Change of coordinates: $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$, $\mathbf{Q} = \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2)$.

$$
\begin{aligned}
\hat{\mathcal{J}}[f(t,\mathbf{p}_1)] = &-\frac{1}{(2\pi)^3\hbar}\int\ldots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{P}'\,\mathrm{d}\mathbf{Q}'\,\delta(\mathbf{P}-\mathbf{P}')\,\delta\left(\frac{\hbar^2}{2m^*}(p_1^2+p_2^2-p_1'^2-p_2'^2)\right) \\
&\times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= &-\frac{1}{(2\pi)^3\hbar}\int\ldots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{Q}'\,\delta\left(\frac{\hbar^2}{4m^*}(4Q^2-4Q'^2)\right) \\
&\times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= &-\frac{1}{(2\pi)^3\hbar}\int\ldots\int \mathrm{d}\mathbf{p}_2 Q'\,\mathrm{d}Q'\,\mathrm{d}\Omega\,\frac{m^*}{\hbar^2}\,\delta(Q^2-Q'^2) \\
&\times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= &-\frac{m^*}{4\pi\hbar^3}\int\ldots\int \frac{\mathrm{d}\mathbf{p}_2}{(2\pi)^2}\int \mathrm{d}\Omega |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right]
\end{aligned}
$$

2.2 Eigenvalue problem

Reducing the Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}}\right) f(t, \mathbf{r}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{r}, \mathbf{p})] \tag{54}$$

- No external force: $\mathbf{F} = \mathbf{0}$,
- No thermal gradient: $\frac{\partial f}{\partial \mathbf{r}} = 0$.

The equation reduces to

$$\frac{\partial f}{\partial t} = \hat{\mathcal{J}}[f] \tag{55}$$

Recall that

- $f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$
- $\delta f(t, \mathbf{p}) = f_0(1 - f_0)\psi(t, \mathbf{p})$
- $\dfrac{\partial f_0}{\partial t} = 0, \quad \hat{\mathcal{J}}[f_0] = 0$

Substituting,

$$\frac{\partial}{\partial t}(f_0(\mathbf{p}) + \delta f(t, \mathbf{p})) = \hat{\mathcal{J}}[f_0(\mathbf{p}) + \delta f(t, \mathbf{p})] \tag{56}$$

$$\frac{\partial}{\partial t} f_0(1 - f_0)\psi(t, \mathbf{p}) = \hat{\mathcal{J}}[f_0(1 - f_0)\psi(t, \mathbf{p})] \tag{57}$$

$$\frac{\partial \psi(t, \mathbf{p})}{\partial t} = \hat{\mathcal{J}}[\psi(t, \mathbf{p})] \tag{58}$$

Proposed ansatz $\psi(t, \mathbf{p}) = \mathrm{e}^{-\gamma t}\phi(\mathbf{p})$ (Chemical potential decays with time)

So, the equation reduces to an eigenvalue problem:

$$\hat{\mathcal{J}}[\phi(\mathbf{p})] = -\gamma\phi(\mathbf{p}) \tag{59}$$

If *perturbation is smooth*, it can be *expanded in the circular harmonics basis*:

$$\mathrm{e}^{im\theta}, \quad m \in \mathbb{Z} \tag{60}$$

For a test mode m,

$$\hat{\mathcal{J}}[\mathrm{e}^{im\theta}] = -\gamma_m \mathrm{e}^{im\theta} \tag{61}$$

Our goal is to study the eigenvalues γ_m as a function of m and temperature T.

3 Odd-Even effect

Trivial solution

The odd-even splitting of the decay rates γ_m derived in this section reproduces the result of Nilsson et al. [1].

$\gamma_m = 0$ is associated with **conserved quantities**.

- $\gamma_0 = 0$
  1. $\phi_0 = 1$: particle number
  2. $\phi_0 = \varepsilon(\mathbf{p})$: energy
- $\gamma_1 = 0$
  1. $\phi_1 = \mathrm{e}^{i\theta}$: x-momentum
  2. $\phi_1 = \mathrm{e}^{-i\theta}$: y-momentum

Near the Fermi surface,

$$\varepsilon(\mathbf{p}_1) \approx \varepsilon(\mathbf{p}_2) \approx \varepsilon(\mathbf{p}_1') \approx \varepsilon(\mathbf{p}_2') \approx \varepsilon_F. \tag{62}$$

Thus, **conservation of energy** implies **conservation of momentum magnitude**:

$$p_1 + p_2 = p_1' + p_2'. \tag{63}$$

Three types of collisions that satisfy this constraint:

- Head-on collisions: $\mathbf{p}_1 = -\mathbf{p}_2, \mathbf{p}_1' = -\mathbf{p}_2'$
- Forward exchange: $\mathbf{p}_1' = \mathbf{p}_2, \mathbf{p}_2' = \mathbf{p}_1$
- No collision: $\mathbf{p}_1' = \mathbf{p}_1, \mathbf{p}_2' = \mathbf{p}_2$

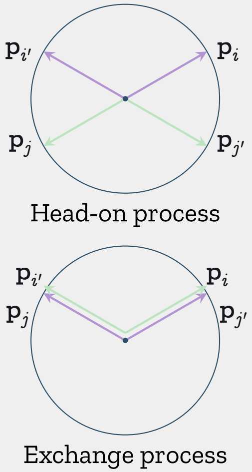

Figure 4: RELEVANT COLLISION TYPES

No collisions: Has no contribution to the collision integral.

Forward exchange & No collisions:

- $\mathbf{p}_1 = \mathbf{p}_1', \mathbf{p}_2 = \mathbf{p}_2'$.
- $\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2') = 0$[3]
- $\therefore$ These collisions has no contribution to the collision integral.

Only **head-on collision** contribute to the collision integral.

[3] Recall that this is a multiplier in the collision integral

Head on collision

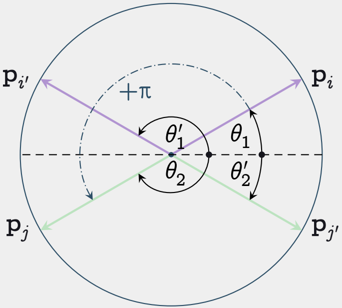

Figure 5: HEAD-ON COLLISION ANGLE

$$\theta_2 = \theta_1 + \pi, \quad \text{and} \quad \theta_2' = \theta_1' + \pi. \tag{64}$$

Substituting test mode m, and recalling $e^{im(\theta+\pi)} = (-1)^m e^{im\theta}$, then

$$\begin{aligned}\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2') \\ = (1 + (-1)^m)e^{im\theta_1} - (1 + (-1)^m)e^{im\theta_{1'}}.\end{aligned} \tag{65}$$

m **is odd:** the term vanishes; thus the collision integral vanishes.

m **is even:** the term does not vanish.

Finding the Even Decay Rate : Overview

- Use **auxiliary variable** $\delta_1 = X - \xi \mathrm{P} \cdot \mathrm{Q}$ and $\delta_2 = X + \xi \mathrm{P} \cdot \mathrm{Q}$, together with $\delta_1 = 0$, and $\theta_{12} = \theta_2 - \theta_1$.
- Collision integral over $\mathbf{p}_2$ and $\mathbf{\Omega}$. Decompose $\mathbf{p}_2$ into **magnitude** p_2 and **angle** θ_2.
- **Change integration variable** from p_2 to δ_2 and θ_2 to $\theta_{12} = \theta_2 - \theta_1$.
- **Omega integral**: Approximate far from $\theta_{12} = 0, \pi$, reduces to standard hyperbolic integral. Gains a T factor.
- **δ_2 integral**: Standard numeric integral. Gains another T factor from change of variables.
- **θ_{12} integral**: Seemingly diverges, but cut off by Ω integral. Gains a $\log(T)$ factor.

Finding the Even Decay Rate : Auxiliary Variables

Recall $X - \xi \mathbf{P} \cdot \mathbf{Q} = \beta(\varepsilon(\mathbf{p}_1) - \mu)$, $X + \xi \mathbf{P} \cdot \mathbf{Q} = \beta(\varepsilon(\mathbf{p}_2) - \mu)$. Define

$$\delta_1 = X - \xi \mathbf{P} \cdot \mathbf{Q}, \quad \text{and} \quad \delta_2 = X + \xi \mathbf{P} \cdot \mathbf{Q}. \tag{66}$$

$$\implies X = \frac{\delta_1 + \delta_2}{2}, \quad \text{and} \quad \xi \mathbf{P} \cdot \mathbf{Q} = \frac{\delta_2 - \delta_1}{2} \tag{67}$$

Assuming $\varepsilon(\mathbf{p}_1) \approx \mu$, $\delta_1 \approx 0$, so $X \approx \xi \mathbf{P} \cdot \mathbf{Q} \approx \dfrac{\delta_2}{2}$.

Finding the Even Decay Rate : Auxiliary Variables

Define $\theta_{12} = \theta_2 - \theta_1$. Then

$$P = 2p_F|\cos(\theta_{12}/2)|, \quad \text{and} \quad Q = p_F|\sin(\theta_{12}/2)|. \tag{68}$$

$$\implies \xi \mathbf{P} \cdot \mathbf{Q}' = \xi PQ \sin \Omega = \xi p_F^2 |\sin(\theta_{12})| \sin \Omega. \tag{69}$$

70

Let $\Lambda = \varepsilon_F \beta = \xi p_F^2$. Finally, $\xi \mathbf{P} \cdot \mathbf{Q}' = \Lambda |\sin(\theta_{12})| \sin \Omega$.

Finding the Even Decay Rate : Integration Variables

The collision integral has the form (note the first (...) contains constants, the second contains dimensionless inteegration quantities like ϕ).

$$(\dots)\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\Omega F_{121'2'}(\dots) \tag{71}$$

Using $\mathrm{d}\mathbf{p}_2 = p_2\,\mathrm{d}p_2\,\mathrm{d}\theta_2, \mathrm{d}\theta_2 = \mathrm{d}\theta_{12}$, and $\mathrm{d}\delta_2 = \beta\,\mathrm{d}\varepsilon(p_2) = \beta\frac{\hbar^2 p_2}{m^*}$, we obtain the form

$$(\dots)\int\dots\int \mathrm{d}\delta_2\,\mathrm{d}\theta_{12}\,\mathrm{d}\Omega F_{121'2'}\cdot(\dots) \tag{72}$$

with constants absorbed into the first (...) (restored later on). Recall $\beta = \frac{1}{k_B T}$. **This is our first T factor**.

Finding the Even Decay Rate : $F_{121'2'}$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh \xi \mathbf{P}\cdot\mathbf{Q}][\cosh X + \cosh \xi \mathbf{P}\cdot\mathbf{Q}']} \tag{73}$$

$$= \frac{1}{4[\cosh(\delta_1+\delta_2)/2 + \cosh(\delta_2-\delta_1)/2][\cosh(\delta_1+\delta_2)/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]} \tag{74}$$

$$= \frac{1}{4[\cosh\delta_2/2 + \cosh\delta_2/2][\cosh\delta_2/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]} \tag{75}$$

$$= \frac{1}{8\cosh\delta_2/2[\cosh\delta_2/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]}. \tag{76}$$

where we have used $\delta_1 = 0$.

Finding the Even Decay Rate : Omega Integral

Collecting terms that depend on $\varOmega$, we have

$$\int_0^{2\pi} \frac{\mathrm{d}\varOmega}{\cosh \delta_2/2 + \cosh\left(\varLambda|\sin(\theta_{12})|\sin \varOmega\right)}. \tag{77}$$

For θ_{12} not near 0 or π, approximate $\sin \varOmega \approx u$, extend the integral to $(-\infty, \infty)$, and change integration variables to $v = \varLambda|\sin(\theta_{12})|u$. Then the integral becomes

$$= \int_{-\infty}^{\infty} \frac{\mathrm{d}\varOmega}{\cosh \delta_2/2 + \cosh\left(\varLambda|\sin(\theta_{12})|\varOmega\right)} = \frac{1}{\varLambda|\sin(\theta_{12})|}\int_{-\infty}^{\infty} \frac{\mathrm{d}v}{\cosh \delta_2/2 + \cosh v} \tag{78}$$

$$= \frac{2\delta_2}{\varLambda|\sin(\theta_{12})|\sinh \delta_2/2} \tag{79}$$

Where we have used the standard integral $\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\cosh a + \cosh x} = \frac{2a}{\sinh a}$. (**Lambda is our second T factor**).

Finding the Even Decay Rate : δ_{12} Integral

Collecting terms that depend on δ_2, we have

$$\frac{1}{8\cosh\delta_2/2}\frac{2\delta_2}{\Lambda\sin(\theta_{12})|\sinh\delta_2/2} = \frac{\delta_2}{2\Lambda\sin(\theta_{12})|\sinh\delta_2}. \quad (80)$$

Using the standard integral $\int_{-\infty}^{\infty}\frac{x}{\sinh x}\,\mathrm{d}x = \frac{\pi^2}{2}$, we have

$$\int_{-\infty}^{\infty}\frac{\delta_2}{2\Lambda\sin(\theta_{12})|\sinh\delta_2}\,\mathrm{d}\delta_2 = \frac{\pi^2}{4\Lambda|\sin(\theta_{12})|}. \quad (81)$$

Finding the Even Decay Rate : θ_{12} Integral

Collecting terms that depend on θ_{12}, we have

$$\int_0^{2\pi} \frac{\mathrm{d}\theta_{12}}{\Lambda|\sin(\theta_{12})|}. \tag{82}$$

The integral is divergent at 0 and π. However, it is cut off by the Ω integral. The cutoff occurs when $\Lambda|\sin(\theta_{12})| \sim 1 \implies |\theta_{12}| \sim T/T_F$. Elsewhere, the antiderivative equals $\ln(\tan(\theta_{12}/2)) + C$.

For θ_{12} at least T/T_F from 0 or π, we have

$$= \int_{T/T_F}^{\pi - T/T_F} \frac{\mathrm{d}\theta_{12}}{\Lambda|\sin(\theta_{12})|}. \tag{83}$$

$$= \frac{1}{\Lambda}[\ln(\tan((\pi - T/T_F)/2)) - \ln(\tan(T/T_F/2))] \tag{84}$$

$$= \frac{1}{\Lambda}[-2\ln(\tan(T/T_F/2))] \tag{85}$$

Using small angle approximation, $\tan(T/T_F/2) \approx T/T_F/2$, we have

$$\int_0^{2\pi} \frac{\mathrm{d}\theta_{12}}{\Lambda|\sin(\theta_{12})|} \approx \frac{2}{\Lambda}\ln(T_F/T). \tag{86}$$

Finding the Even Decay Rate : Final Result

Collecting all the T factors, we have

- One from β in the change of variables from p_2 to δ_2.
- One from the Λ from Ω integral.
- One log factor from the θ_{12} integral.

Combining constants obtained during the change of integration variable, we obtain

Collision Integral Approximation to Zero-th order

$$-\gamma_m \phi = \hat{\mathcal{J}}[\phi] = -\frac{k_B \pi^2 m^{*2} |V|^2}{\hbar^4 (2\pi)^3} \frac{T^2}{T_F} \ln(\frac{T_F}{T}) \phi \tag{87}$$

Thus, we have $\gamma_m \propto T^2 \log T$. Note that this only works for small ϕ.

Strategy for the Odd Decay Rate

Since, to the zeroth order, the ϕ terms vanish, we need the second order perturbation (first order also vanishes). The strategy is to

- Develop eigenvalue perturbation around $\delta_T = T/T_F \ll 1$
- Expand the ϕ around 0 to obtain a power series in $\varepsilon(p_i)/\varepsilon_F = \delta_T u_i$, where $u_i = \varepsilon_i/T$.
- Integrate the power series against the Fermi functions.
- Use the eigenvalue perturbation theory to obtain the lowest order approximation of γ_m.

4 Further works

van Hove singularity

van Hove singularity: An **energy level** that has infinitely many available states.[a]

[a] Spike in the density of states

- More available states $\implies$ stronger pairing interactions (forming Cooper pairs)
- Superconductivity can happen

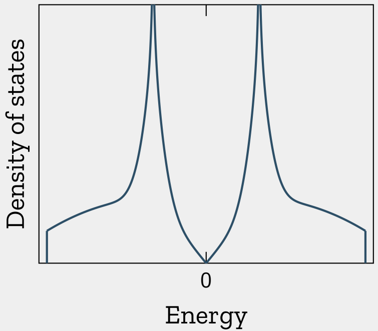

Figure 6: VAN-HOVE SINGULARITY (NOT TO SCALE, NOT REAL DATA)

If Fermi surface has the **energy that's close to the van Hove singularity**, superconductivity can occur more easily.

- Does the odd-even effect persists near the van Hove singularity?
- What's the relaxation time?

These questions extend the hierarchy of long-lived modes described in Ledwith et al. [2] and the odd-even splitting of Nilsson et al. [1] beyond the simple parabolic-band Fermi surface.

Bibliography

[1] E. Nilsson, U. Gran, and J. Hofmann, Phys. Rev. X **15**, 041007 (2025).

[2] P. J. Ledwith, H. Guo, and L. Levitov, Annals of Physics **411**, 167913 (2019).

Thank you!